

Real-time fare-deal intelligence for the MENAT+CA region

1. Executive summary

The MENAT+CA region — roughly 600 million people, four global mega-hubs, and a foreign-resident workforce that flies internationally several times a year — is materially underserved by deal-detection products. Going, Hopper, and Jack's Flight Club are anchored on US/EU origins. Wego, Almosafer, and Cleartrip operate as price-aggregators rather than deal-detectors. The opportunity is a carrier-aware, baseline-driven, real-time fare anomaly detection system built on first-party API data, designed for the price volatility, holiday cycles, and currency dynamics specific to this region.

We propose a six-phase build over 12–14 weeks producing a production-grade monitoring stack with verified phantom-fare filtering, region-specific seasonal modelling, and a dashboard suitable for both internal operations and B2C alerting.

2. Market opportunity

Driver	What it means for fare-deal monitoring
Four super-connector hubs (IST, DXB, DOH, AUH)	One origin can be alerted to ~150 destinations via a single carrier; pricing volatility on hub-to-spoke routes creates frequent deal windows
Large expat / VFR traffic (Gulf South Asia, Turkey CIS, North Africa Europe)	Predictable seasonal demand patterns make baseline modelling tractable
Pronounced holiday spikes: Ramadan, Eid al-Fitr, Eid al-Adha, Hajj/Umrah, Nowruz, Turkish school holidays	Creates strong seasonality features — not noise, but signal
Multi-currency pricing (AED, SAR, QAR, TRY, EGP, KZT, UZS) with FX volatility	Currency normalisation is a competitive moat; competitors do this poorly
Regional carriers underrepresented in Western tools	Air Astana, Uzbekistan Airways, Pegasus, Royal Air Maroc rarely surface in Going / Hopper alerts

Competitive positioning: not a metasearch, not an OTA. A deal-detection engine with verified bookable prices that can be productised as (a) a B2C alert app, (b) a B2B feed for OTAs, or (c) an internal pricing-intelligence tool.

3. Product definition

- Unit of analysis: route-cabin-date-carrier observation, e.g. ISTALA, business, 2026-09-14, Turkish Airlines.
- "Deal" trigger: current bookable price X% below the rolling baseline for the same (origin, dest, cabin, days-to-departure bucket, season) quadruple, where X is configurable per cabin (suggested defaults: 18% economy, 25% premium economy, 30% business).
- Latency target: detection within one polling cycle (10 min), verification within 60 seconds, alert delivery within 2 minutes thereafter.
- Verification: every alert is re-priced via a secondary API call before send, eliminating phantom fares.
- Output channels: Telegram (Phase 3 default), email, Slack/webhook (Phase 5), iOS push (out of scope unless productised).

4. Geographic and carrier coverage

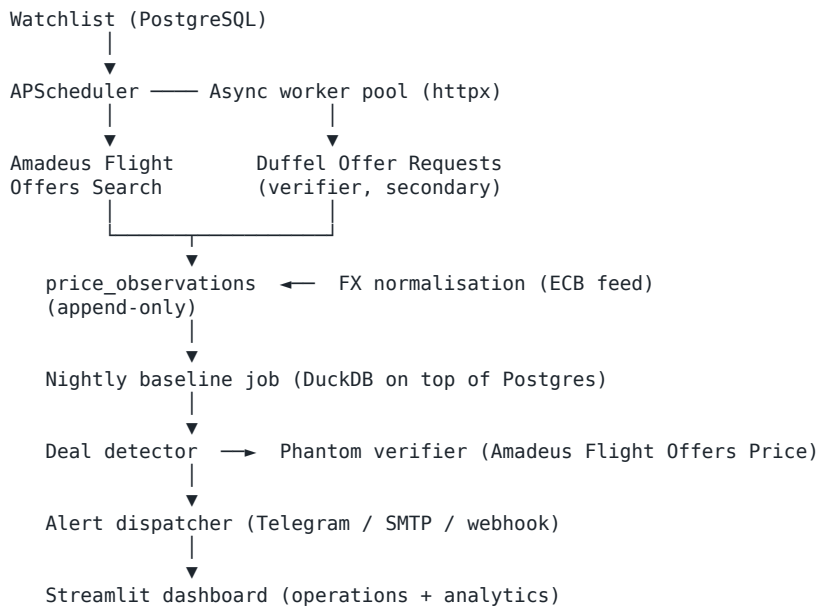
Origin clusters (Phase 1 = bold):

- Gulf: DXB, AUH, DOH, RUH, JED, BAH, KWI, MCT
- Turkey: IST, SAW
- Levant: AMM, BEY, TLV (handled with operational care)
- North Africa: CAI, CMN, RAK, TUN, ALG
- South Caucasus: GYD, EVN, TBS
- Central Asia: ALA, TSE, TAS, FRU, DYU
- Iraq/Iran: BGW excluded politically; IKA excluded under sanctions

Carrier tiers:

Tier	Coverage in Phase 1	Carriers
1 — Anchor	Yes	Emirates, Qatar Airways, Turkish Airlines, Etihad, Saudia
2 — Regional majors	Yes	Royal Jordanian, EgyptAir, Royal Air Maroc, Oman Air, Gulf Air, Kuwait Airways, Air Astana, Uzbekistan Airways, Azerbaijan Airlines
3 — LCC & secondary	Phase 2	flydubai, Air Arabia, Pegasus, AnadoluJet, FlyArystan, SCAT, Jazeera, Salam Air, Wizz Air Abu Dhabi
4 — Niche / spotty distribution	Phase 5 if at all	Avia Traffic, Air Manas, Somon Air, Iraqi Airways, Tunisair, Air Algérie, Buta, Cyprus Airways

5. Technical architecture



Stack: Python 3.12, PostgreSQL 16, Amadeus Self-Service API (primary), Duffel (secondary verifier), DuckDB for analytical queries, APScheduler, Streamlit, Docker, Telegram Bot API.

Why this architecture:

- Append-only observations. Every API response stored as a row, never overwritten. Baseline integrity depends on this.
- Two-API verification. Amadeus for primary search and price-confirm; Duffel as a free secondary because its search is unbilled.
- DuckDB on top of Postgres for analytics. Postgres is the system-of-record; DuckDB runs the heavy baseline windowing without taxing the OLTP path.
- Region-aware seasonality. Hijri-calendar holidays (Ramadan, Eid, Hajj) and regional school terms encoded as feature flags in the baseline model. Western tools don't do this.

6. Data sources — what we use and why

Source	Role	Status (verified 2026-05-10)
Amadeus Self-Service API	Primary search + final price-confirm	Free monthly quota carries to production; ~1 req/100ms in test
Duffel API	Secondary verifier (free; only bookings billed)	Live search rate-limit 120 req/min; airline coverage limited at signup
ECB FX reference rates	Currency normalisation to GBP/EUR	Free, daily
Hijri calendar feed	Ramadan/Eid/Hajj feature flags	Static lookup, refreshed annually
Carrier direct sites	Excluded — ToS forbids automated access	CJEU Ryanair v PR Aviation upholds enforceability
Wego / Almosafer / Skyscanner scraping	Excluded — ToS forbids	—
Kiwi Tequila	Excluded — invite-only since May 2024	—

7. Roadmap and milestones

Phase	Weeks	Deliverable	Definition of done
0 — Spec & decisions	1	Locked spec doc; Amadeus + Duffel accounts provisioned; route watchlist v1	One-page approved spec; sandbox API keys live
1 — Data foundation	2–3	Amadeus client, watchlist table, observations log, scheduler running	4 weeks of clean observations on 30 anchor routes; row-count and dedupe checks passing
2 — Baselines & detection	2	Baseline job, deal detector, phantom verifier, backtest harness	Backtest report: precision/recall on synthetic deal injections; per-route baseline health dashboard
3 — Alerts	1	Telegram bot, alert formatter with deep links + baseline context	First end-to-end alert delivered to phone within latency target
4 — Dashboard	1	Streamlit ops panel: watchlist, observations, baseline health, recent alerts	Internal user can audit any alert end-to-end without touching the database
5 — Productisation	2–3	LCC tier added, multi-user auth, webhook output, FX normalisation hardened	B2B partner can consume the alert feed
6 — Optional: award seats	2+	Carrier-specific award scrapers where ToS permits via affiliate APIs	Out of scope unless commercial case approved

Total committed scope: phases 0–5, ~12 weeks.

8. KPIs and success metrics

Metric	Definition	Target
Alert precision	% of alerts that remain bookable at the alerted price for 30 min after delivery	85%
Alert recall (sampled)	% of true price drops on watched routes that triggered an alert	70% on Tier 1 carriers
Phantom rejection rate	% of detector hits filtered out by the verifier	Tracked, expected 15–25%
Mean detect-to-alert latency	Time from price drop appearing in our DB to alert delivery	4 min p95
Baseline health	% of watched routes with 30 observations in current bucket	90% by end of Phase 2
API spend per alert	Amadeus + Duffel cost / number of alerts delivered	Tracked weekly

9. Risk register

Risk	Likelihood	Impact	Mitigation
Amadeus quota exhaustion	Medium	High	Per-endpoint budget tracker; fallback to lower poll cadence; alert on quota burn rate
Phantom fares slip past verifier	Low-Medium	Medium	Two-strike rule (must appear in 2 consecutive fetches); user-facing confidence flag
Region-specific seasonality (Ramadan, Hajj) blows out baselines	High in first cycle	Medium	Hijri-calendar feature flag; first cycle treated as bootstrap not production
Carrier coverage gaps in LCC tier	High	Low-Medium	Honest "not monitored" status; phased addition in Phase 2
FX volatility mistaken for price drops	Medium	Medium	Store native currency; convert at display only; alert thresholds applied in native currency
Mistake fare cancelled by airline	Medium	Reputational	Alert metadata flags suspected mistake fares; user advisory included; UK common-law unilateral-mistake doctrine documented
Sanctions exposure (IR, SY, RU corridors)	Low	High	Hard exclusion list at the watchlist layer; documented compliance posture
Personal data / GDPR (UK + EU + Turkish KVKK + UAE PDPL)	Medium	High	Minimal PII; encrypted at rest; per-jurisdiction DPA before any user data is processed

10. Resource requirements

- One developer (the proposer), full-stack Python, SQL, scraping experience — already operating an Amadeus PostgreSQL Streamlit stack, transferable directly.
- Infrastructure: single VPS sufficient for Phases 1–4 (4 vCPU / 16 GB RAM / 200 GB SSD £25/month). Phase 5 may require horizontal split.
- API spend: estimated £30–80/month during Phase 1 (Amadeus + Duffel sandbox), scaling to £150–300/month at Phase 5 production load. Exact figure depends on watchlist size — modelled in build prompt #3.
- External: legal review before any commercial / B2B deployment (DPA + carrier ToS for redistribution rights).